Quantitative Comparisons

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Numeric comparisons**

Column A: 7×8. Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

$A = 56$. $B = 54 + 2 = 56$. $A = B$.

2 **[Foundational]** • **Target: Numeric comparisons**

Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.

$A = 579$. $B = 580$. $B > A$.

3 **[Foundational]** • **Target: Percent comparisons**

Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

$A = 0.25 \times 160 = 40$. $B = 0.20 \times 200 = 40$. $A = B$.

4 **[Foundational]** • **Target: Percent comparisons**

Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

$A = 100 \times 1.1 \times 1.1 = 121$. $B = 100 \times 1.2 = 120$. $A > B$.

5 **[Foundational]** • **Target: Algebraic comparisons at a fixed value**

Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

$A = 2(5) + 3 = 13$. $B = 5 + 8 = 13$. $A = B$.

6 **[Foundational]** • **Target: Comparisons requiring testing values**

For any real number x: Column A: x. Column B: x^2. Compare the two columns.

If $x = 2$: $B > A$. If $x = \frac{1}{2}$: $A > B$. If $x = 1$: $A = B$. The relationship changes, so it cannot be determined.

7 **[Foundational]** • **Target: Comparisons requiring testing values**

Given only that n is an integer: Column A: n^2. Column B: 4. Compare the two columns.

If $n = 1$: $A = 1 < 4$. If $n = 3$: $A = 9 > 4$. The relationship changes, so it cannot be determined.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Geometry comparisons**

Column A: the perimeter of a square with side 6 cm. Column B: the perimeter of a rectangle with length 8 cm and width 3 cm. Compare the two columns.

$A = 4 \times 6 = 24$ cm. $B = 2(8 + 3) = 22$ cm. $A > B$.

9 **[Intermediate]** • **Target: Geometry comparisons**

Column A: the area of a circle with radius 4 cm (use $\pi \approx 3.14$). Column B: the area of a square with side 7 cm. Compare the two columns.

$A = 3.14 \times 4^2 = 3.14 \times 16 = 50.24$ cm^2. $B = 7^2 = 49$ cm^2. $A > B$.

10 **[Intermediate]** • **Target: Geometry comparisons**

Column A: the sum of the interior angles of a pentagon. Column B: the sum of the interior angles of two triangles. Compare the two columns.

$A = (5 - 2) \times 180^\circ = 540^\circ$. $B = 2 \times 180^\circ = 360^\circ$. $A > B$.

11 **[Intermediate]** • **Target: Geometry comparisons**

Column A: the hypotenuse of a right triangle with legs 5 and 12. Column B: 14. Compare the two columns.

$A = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$. $B = 14$. $B > A$.

12 **[Intermediate]** • **Target: Ratio and proportion comparisons**

Column A: the value of x, if $x : 12 = 3 : 4$. Column B: 10. Compare the two columns.

$x = \frac{3 \times 12}{4} = 9$. $B = 10$. $B > A$.

13 **[Intermediate]** • **Target: Ratio and proportion comparisons**

A recipe uses flour and sugar in the ratio $3 : 1$. Column A: the sugar needed for 240 g of flour. Column B: 90 g. Compare the two columns.

Sugar $= 240 \div 3 = 80$ g. $B = 90$ g. $B > A$.

14 **[Intermediate]** • **Target: Ratio and proportion comparisons**

Column A: the sum of the parts in the simplified form of the ratio $18 : 24$. Column B: 8. Compare the two columns.

$18 : 24$ simplifies to $3 : 4$, so the sum of parts is $3 + 4 = 7$. $B = 8$. $B > A$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 3.01 Algebra Essentials

Two numbers are in the ratio $5:3$, and their sum is 96. Column A: the larger number. Column B: 61. Compare the two columns.

Total parts $= 5 + 3 = 8$, each part $= 12$. Larger number $= 5 \times 12 = 60$. $B = 61$. $B > A$.

16 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 4.01 Geometry & Measurement

A data set is $4, 7, 7, 9, 13$. Column A: the mean. Column B: the median. Compare the two columns.

Mean $= \frac{4+7+7+9+13}{5} = \frac{40}{5} = 8$. Median (middle value) $= 7$. $A > B$.

17 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 6.01 Sequences & Patterns

Column A: the range of the data set $3, 8, 15, 22$. Column B: 20. Compare the two columns.

Range $= 22 - 3 = 19$. $B = 20$. $B > A$.

18 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

A bag contains 3 red balls and 5 blue balls. Column A: the probability of drawing a red ball. Column B: the probability of drawing a blue ball. Compare the two columns.

$P(\text{red}) = \frac{3}{8}$. $P(\text{blue}) = \frac{5}{8}$. $B > A$.

19 [Challenging] • Target: Word-problem comparisons • Cross-Lesson: 8.01 Word Problems & Reasoning

Column A: the total cost of 4 items at 15 riyals each, after a 10% discount on the total. Column B: 55 riyals. Compare the two columns.

Total before discount $= 4 \times 15 = 60$ riyals. After discount: $60 \times 0.9 = 54$ riyals. $B = 55$. $B > A$.

20 [Challenging] • Target: Word-problem comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

Ali's age is 3 years more than twice Sara's age. Sara is 8. Column A: Ali's age. Column B: 20. Compare the two columns.

Ali $= 2(8) + 3 = 19$. $B = 20$. $B > A$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Percent comparisons

Column A: the result of decreasing 100 by 20%, then increasing that result by 20%. Column B: 96.

a. The relationship cannot be determined from the given information

b. **The two columns are equal ✓**

c. Column A is greater

d. Column B is greater

$100 \times 0.8 \times 1.2 = 96$: equal.

22 [Foundational] • Target: MCQ — Comparisons requiring testing values

For any integer n: Column A: n. Column B: $-n$. Compare the two columns.

a. The two columns are equal

b. **The relationship cannot be determined from the given information ✓**

c. Column B is greater

d. Column A is greater

If $n > 0$, A is greater; if $n < 0$, B is greater; if $n = 0$, they are equal — it varies.

23 [Foundational] • Target: MCQ — Comparisons requiring testing values

Given only that m is a positive integer: Column A: m. Column B: m^2. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. **The relationship cannot be determined from the given information ✓**

At $m = 1$ they are equal; for $m > 1$, Column B is greater — it varies.

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24 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. **Column B is greater ✓**

$C = 2(3.14)(5) = 31.4$, which is less than 32.

25 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column B is greater

d. **Column A is greater ✓**

Rectangle area $= 9 \times 4 = 36$; square perimeter $= 4 \times 7 = 28$. Column A is greater.

26 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the sum of the interior angles of a hexagon. Column B: the sum of the interior angles of two triangles.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. **Column A is greater ✓**

d. Column B is greater

Hexagon $= 720^\circ$; two triangles $= 180 + 180 = 360^\circ$. Column A is greater.

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27 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the hypotenuse of the right triangle shown. Column B: 16.

a. The relationship cannot be determined from the given information

b. **Column B is greater ✓**

c. The two columns are equal

d. Column A is greater

$\sqrt{9^2 + 12^2} = \sqrt{225} = 15$, which is less than 16.

28 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. **Column B is greater ✓**

$V = 3.14 \times 9 \times 7 = 197.82$, which is less than 200.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Ratio and proportion comparisons

Column A: the value of x, if $x : 15 = 2 : 5$. Column B: 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. **Column B is greater ✓**

$x = 15 \times \frac{2}{5} = 6$, which is less than 7.

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30 **[Intermediate]** • **Target: MCQ — Ratio and proportion comparisons**

Two numbers are in the ratio $4:7$, and their sum is 99. Column A: the smaller number. Column B: 40.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **Column B is greater ✓**

Each part $= 99 \div 11 = 9$; smaller $= 4 \times 9 = 36$, which is less than 40.

31 **[Intermediate]** • **Target: MCQ — Ratio and proportion comparisons**

Column A: the sum of the parts in the simplified form of the ratio $24:36$. Column B: 5.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **The two columns are equal ✓**

$24:36$ simplifies to $2:3$; $2 + 3 = 5$: equal.

32 **[Intermediate]** • **Target: MCQ — Ratio and proportion comparisons**

A recipe uses water and rice in the ratio $2:1$. Column A: the water needed for 300 g of rice. Column B: 550 g.

a. Column B is greater

b. **Column A is greater ✓**

c. The two columns are equal

d. The relationship cannot be determined from the given information

Water $= 2 \times 300 = 600$ g, which is greater than 550 g.

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33 [Intermediate] • Target: MCQ — Data and probability comparisons

A data set is $5, 8, 8, 10, 14$. Column A: the mean. Column B: the median.

a. The two columns are equal

b. Column B is greater

c. **Column A is greater ✓**

d. The relationship cannot be determined from the given information

Mean $= \frac{45}{5} = 9$; median $= 8$. Column A is greater.

34 [Intermediate] • Target: MCQ — Data and probability comparisons

Column A: the range of the data set $6, 11, 19, 25$. Column B: 19.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. **The two columns are equal ✓**

d. Column A is greater

Range $= 25 - 6 = 19$: equal.

35 [Intermediate] • Target: MCQ — Data and probability comparisons

A bag contains 4 green balls and 9 yellow balls. Column A: the probability of drawing a green ball. Column B: 0.4.

a. **Column B is greater ✓**

b. The relationship cannot be determined from the given information

c. The two columns are equal

d. Column A is greater

$P = \frac{4}{13} \approx 0.31$, which is less than 0.4: Column B is greater.

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36 [Intermediate] • Target: MCQ — Data and probability comparisons

A data set is $2, 4, 4, 4, 6$. Column A: the mode. Column B: the mean.

a. Column A is greater

b. The relationship cannot be determined from the given information

c. **The two columns are equal ✓**

d. Column B is greater

Mode $= 4$; mean $= \frac{20}{5} = 4$: equal.

37 [Intermediate] • Target: MCQ — Word-problem comparisons

Column A: the total cost of 5 items at 20 riyals each, after a 15% discount on the total. Column B: 90.

a. The two columns are equal

b. Column A is greater

c. The relationship cannot be determined from the given information

d. **Column B is greater ✓**

$5 \times 20 = 100$; after discount $= 100 \times 0.85 = 85$, which is less than 90.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Word-problem comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

Layla's age is 4 years less than three times Huda's age. Huda is 9. Column A: Layla's age. Column B: 25.

a. Column A is greater

b. The two columns are equal

c. **Column B is greater ✓**

d. The relationship cannot be determined from the given information

Layla $= 3(9) - 4 = 23$, which is less than 25.

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39 [Challenging] • Target: MCQ — Word-problem comparisons • Cross-Lesson: 8.01 Word Problems & Reasoning

Column A: the simple interest on 3,000 riyals at 4% per year for 2 years. Column B: 250.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **Column B is greater ✓**

$I = 3{,}000 \times 0.04 \times 2 = 240$, which is less than 250.

40 [Challenging] • Target: MCQ — Word-problem comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

A car travels 300 km using 25 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km.

a. **The two columns are equal ✓**

b. The relationship cannot be determined from the given information

c. Column B is greater

d. Column A is greater

$300 \div 25 = 12$ km per liter: equal.

41 [Challenging] • Target: MCQ — Comparisons with inequalities and ranges • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Given that $3 < x < 8$. Column A: x. Column B: 5.

a. **The relationship cannot be determined from the given information ✓**

b. The two columns are equal

c. Column A is greater

d. Column B is greater

x could be less than, equal to, or greater than 5 within this range — it varies.

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42 [Challenging] • Target: MCQ — More geometry comparisons • Cross-Lesson: 3.01 Algebra Essentials

Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. **Column B is greater ✓**

d. The two columns are equal

Cube $= 5^3 = 125$; box $= 4 \times 5 \times 7 = 140$. Column B is greater.

43 [Challenging] • Target: MCQ — More geometry comparisons • Cross-Lesson: 4.01 Geometry & Measurement

Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the area of a square with side 11 cm.

a. The two columns are equal

b. Column A is greater

c. **Column B is greater ✓**

d. The relationship cannot be determined from the given information

Circumference $= 2 \times \frac{22}{7} \times 7 = 44$; square area $= 121$. Column B is greater.

44 [Challenging] • Target: MCQ — More geometry comparisons • Cross-Lesson: 6.01 Sequences & Patterns

Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.

a. Column B is greater

b. The two columns are equal

c. **Column A is greater ✓**

d. The relationship cannot be determined from the given information

Octagon $= (8 - 2) \times 180 = 1{,}080^\circ$; four triangles $= 4 \times 180 = 720^\circ$. Column A is greater.

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45 [Challenging] • Target: Numeric comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

For any real x. Column A: x^2. Column B: 4. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. **The relationship cannot be determined from the given information ✓**

d. The two columns are equal

x^2 can be less than, equal to, or greater than 4 depending on x (e.g. $x = 0$ gives $x^2 = 0$; large x gives $x^2 > 4$), so the relationship is not constant.

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